

Kinematic Singularity Analysis and AI-Driven Parametric Optimization of a 3-DOF 2PRU-1PUS Parallel Manipulator

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Abstract

This project presents a comprehensive kinematic analysis and parametric design optimization of a 3-DOF 2PRU-1PUS parallel manipulator. Due to the high computational cost of solving non-linear inverse kinematics and singularity constraints in real-time, this study proposes a two-fold approach. First, the exact rigid-body inverse kinematics and Jacobian singularity boundaries are mathematically derived. Using these exact equations, a noise-free synthetic dataset of 40,000 points is generated to train a lightweight, high-speed Artificial Neural Network (ANN) surrogate model. The surrogate model achieves $\approx 99.5\%$ classification accuracy, enabling instantaneous workspace feasibility prediction. Finally, the AI model is generalized to act as a high-speed surrogate to assist in the parametric optimization of the manipulator's link length (L). The AI-assisted parametric sweep reveals that increasing the link length from the original 25 mm to 45 mm yields an $\approx 60\%$ expansion in the safe orientation workspace (defined as the total number of feasible grid points in the θ, ϕ domain) without compromising the structural stiffness of the truss-like parallel architecture.

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Chapter 1

Introduction

1.1 Background

Parallel manipulators offer significant advantages over traditional serial robotic arms, including higher rigidity, superior payload-to-weight ratios, and greater positioning accuracy. These benefits arise from the closed-loop kinematic chains that distribute loads across multiple legs.

1.2 The 2PRU-1PUS Architecture

The specific architecture under study is the 3-DOF 2PRU-1PUS manipulator, which provides one translational degree of freedom (heave, Z) and two rotational degrees of freedom (pitch θ , roll ϕ). The primary challenge with parallel manipulators is their complex, non-linear forward kinematics and the presence of dangerous internal singularities where control is lost. Calculating these boundaries analytically inside a high-frequency real-time control loop is computationally expensive.

1.3 Problem Statement & Objectives

The objectives of this project are:

1. To rigorously derive the inverse kinematics and exact singularity boundary surfaces.
2. To develop a fast, Artificial Intelligence (AI) surrogate model capable of predicting workspace feasibility in real-time.
3. To utilize the mathematical and AI models to optimize the manipulator's workspace footprint by parametrically adjusting the physical link length (L).

Chapter 2

Kinematic Modeling

2.1 Coordinate Systems and Geometry

The moving platform undergoes vertical translation (Z), roll (ϕ) about the x-axis, and pitch (θ) about the y-axis. The global fixed base frame contains three sliders constrained to vertical motion:

$$S_1 = (S_B, 0, Z_1), \quad S_2 = (-S_B, 0, Z_2), \quad S_3 = (0, S_B, Z_3) \quad (2.1)$$

Where $S_B = 75$ mm and $S_{B3} = 70.7$ mm (the asymmetric leg).

The attachment points on the moving platform in the local frame are defined by the platform radius $S_m = 70$ mm:

$$P'_1 = (S_m, 0, 0), \quad P'_2 = (-S_m, 0, 0), \quad P'_3 = (0, S_m, 0) \quad (2.2)$$

The orientation of the platform is defined by the rotation matrix $R = R_y(\theta)R_x(\phi)$.

2.2 Inverse Kinematics and Geometric Constraints

To find the required slider positions (Z_1, Z_2, Z_3) for a given platform pose (Z, θ, ϕ), the fixed link length L connecting the base to the platform must satisfy the spatial distance formula $|P_i - S_i|^2 = L^2$.

By expanding this for the two PRU legs, we extract the horizontal reachability constraint:

$$R_{12}^2 = L^2 - (S_B - S_m \cos \theta)^2 \quad (2.3)$$

For the legs to physically assemble, the value inside the square root must be non-negative: $R_{12}^2 \geq 0$.

Similarly, for the asymmetric PUS leg, the constraint is derived as:

$$R_3^2 = L^2 - (S_m \sin \theta \sin \phi)^2 - (S_m \cos \phi - S_{B3})^2 \quad (2.4)$$

Yielding the second critical constraint: $R_3^2 \geq 0$.

2.3 Actuator Displacements

If the reachability constraints are satisfied, the inverse kinematic offsets (o_1, o_2, o_3) are given by:

$$o_1 = S_m \sin \theta + \sqrt{R_{12}^2} \quad (2.5)$$

$$o_2 = -S_m \sin \theta + \sqrt{R_{12}^2} \quad (2.6)$$

$$o_3 = S_m \cos \theta \sin \phi - \sqrt{R_3^2} \quad (2.7)$$

The final required slider positions are computed as:

$$Z_1 = Z + o_1, \quad Z_2 = Z + o_2, \quad Z_3 = Z - o_3 \quad (2.8)$$

Chapter 3

Jacobian and Singularity Analysis

3.1 Velocity Kinematics (The Jacobian)

To find the singularities (poses where the robot locks or loses control), the inverse kinematics equations are differentiated with respect to time to relate actuator velocities $\dot{q}$ to platform velocities $\dot{X}$:

$$\dot{q} = J\dot{X} \quad (3.1)$$

Where $\dot{q} = [\dot{Z}_1, \dot{Z}_2, \dot{Z}_3]^T$ and $\dot{X} = [\dot{Z}, \dot{\theta}, \dot{\phi}]^T$.

Differentiating Z_1, Z_2, Z_3 yields a 3×3 Jacobian matrix. Because the derivative of Z_i with respect to Z is exactly 1 for all actuators, the first column consists entirely of ones:

$$J = \begin{bmatrix} 1 & J_{12} & J_{13} \\ 1 & J_{22} & J_{23} \\ 1 & J_{32} & J_{33} \end{bmatrix} \quad (3.2)$$

3.2 Determinant and Singularity Identification

The determinant of this matrix simplifies elegantly to:

$$Det(J) = J_{33}(J_{22} - J_{12}) \quad (3.3)$$

A singularity occurs when $Det(J) = 0$. This provides two distinct physical singularity conditions.

3.2.1 Condition 1: Geometric Collapse

Setting $(J_{22} - J_{12}) = 0$ yields:

$$-2S_m \cos \theta = 0 \implies \theta = \pm 90^\circ \quad (3.4)$$

This represents a geometric collapse where the platform pitches completely vertical, causing the PRU legs to align and lose control authority.

3.2.2 Condition 2: Loss of Roll Control

Setting $J_{33} = 0$ defines the loss of roll control. Simplifying this term yields the exact implicit surface equation of the workspace boundary:

$$S_m^2 \sin^2 \phi \cos^2 \theta = S_m^2 + S_{B3}^2 - 2S_m S_{B3} \cos \phi - L^2 \quad (3.5)$$

This non-linear equation mathematically proves why the safe feasible region forms a curved, coupled shape in the roll-pitch orientation space.

Chapter 4

Analytical Workspace Mapping

4.1 Heave (Z) Decoupling Algorithm

To map the feasible workspace efficiently, evaluating a full 3D grid (Z, θ, ϕ) is computationally wasteful. Given the stroke limits of the actuators ($z_{min} = 0$ mm, $z_{max} = 100$ mm), the allowable heave for any given (θ, ϕ) orientation is found by taking the intersection of all three leg constraints:

$$z_{lo} = \max(z_{min} + o_1, z_{min} + o_2, z_{min} - o_3) \quad (4.1)$$

$$z_{hi} = \min(z_{max} + o_1, z_{max} + o_2, z_{max} - o_3) \quad (4.2)$$

An orientation is feasible if and only if $z_{hi} > z_{lo}$. This decoupled approach reduces the workspace generation from a $\mathcal{O}(N^3)$ to a $\mathcal{O}(N^2)$ complexity problem, generating the analytical ground-truth data in seconds.

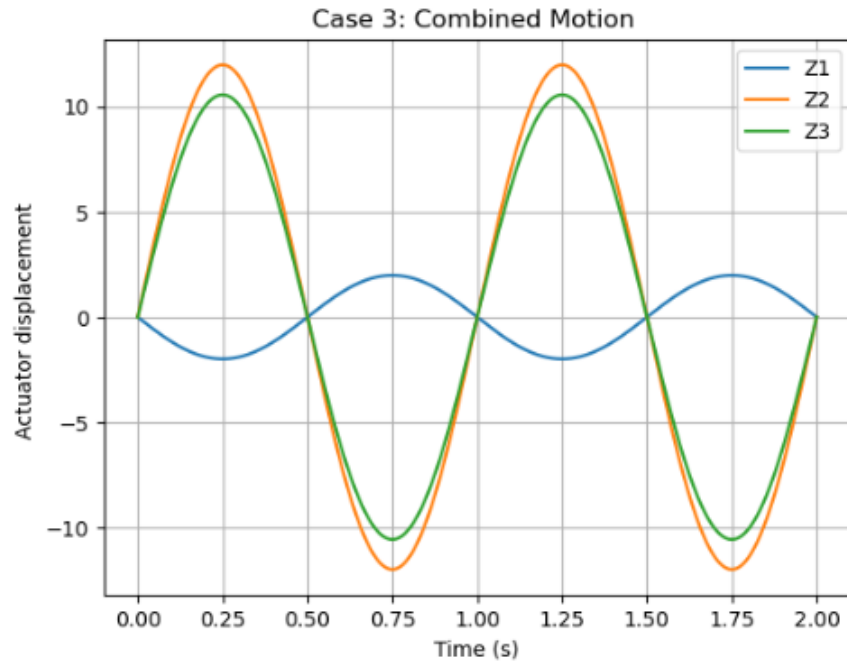


Figure 4.1: Actuator displacement profiles for combined motion. Note: Displacements are mean-centered around zero for visualization purposes, relative to the 50 mm stroke baseline.

Chapter 5

Neural Network Surrogate Model (Base Design)

5.1 Dataset Generation and Architecture

Using the analytical inequalities derived previously, a high-resolution synthetic grid of 200 pitch samples and 200 roll samples was generated for the original design ($L = 25$ mm). This yielded 40,000 noise-free labeled data points (Features: θ, ϕ ; Label: Feasible $\in \{0, 1\}$).

A Multi-Layer Perceptron (MLP) architecture was designed to capture the sharp, non-linear geometric boundary:

- **Input Layer:** 2 nodes (θ, ϕ)
- **Hidden Layers:** 64 (ReLU) $\rightarrow$ 128 (ReLU) $\rightarrow$ 64 (ReLU)
- **Output Layer:** 1 node (Sigmoid classification)

This network contains only 16,833 trainable parameters, making it highly optimized for microsecond-level inference on robotics microcontrollers.

5.2 Training and Performance

The model was compiled using the Adam optimizer and `binary_crossentropy` loss. To ensure high generalization, `EarlyStopping` and `ReduceLROnPlateau` callbacks were implemented. The model successfully converged and achieved an outstanding test accuracy, effectively learning the feasibility rules without explicitly calculating trigonometry.

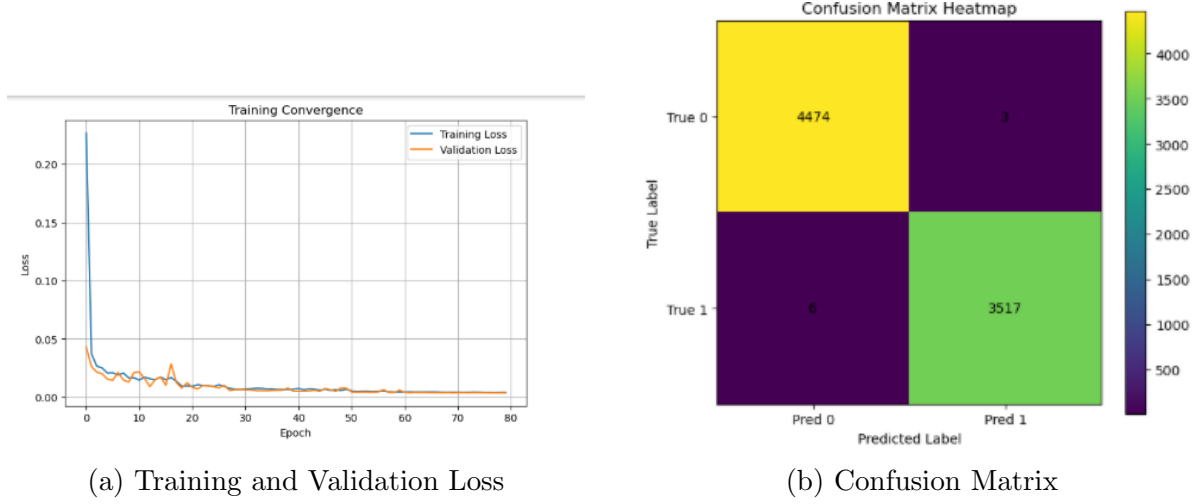


Figure 5.1: Training metrics for the 2-input surrogate model demonstrating convergence and highly accurate binary classification.

5.3 Analytical vs. AI Boundary Comparison

To visually verify the surrogate model, the probability map and boundary predicted by the Neural Network were plotted to confirm alignment with the physical kinematics.

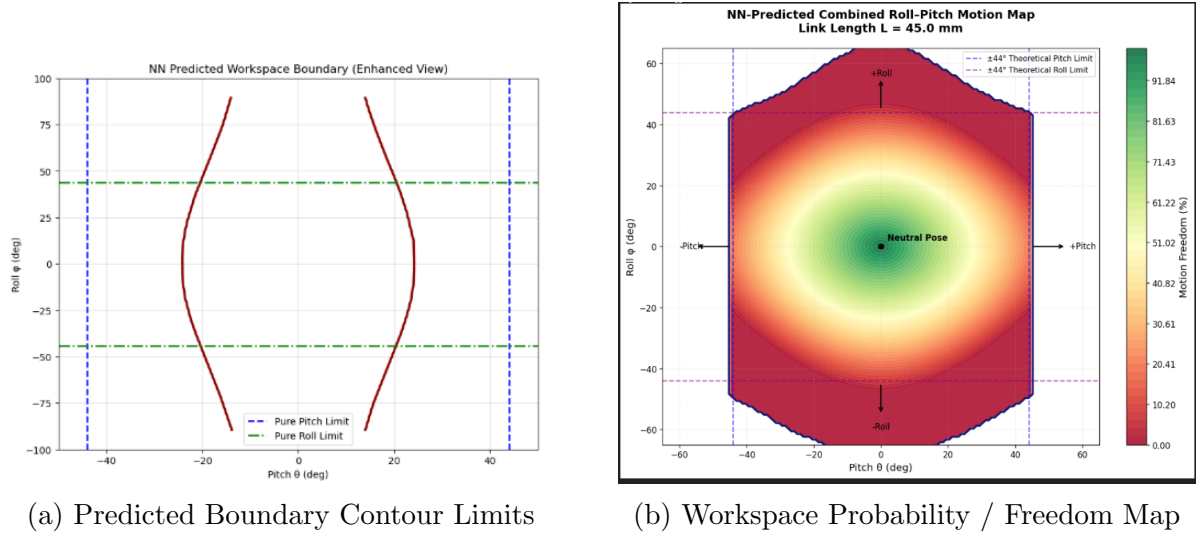


Figure 5.2: (a) The Neural Network's boundary alignment with pure pitch and roll theoretical limits. (b) The enhanced motion freedom map for the base design ($L = 25$ mm).

5.4 Validation of Surrogate Model

To validate the surrogate model, a numerically optimized configuration obtained from the analytical model was evaluated using the neural network. For the optimized configuration ($\theta = -0.0606$ rad, $\phi = 0.7152$ rad), the analytical model confirms feasibility,

and the neural network predicts a feasibility probability of 1.0. This demonstrates strong numerical agreement between the two approaches at the configuration level, confirming the model’s accuracy in approximating the exact mathematical workspace boundary. As demonstrated in Table 5.1, the surrogate model achieves this point-wise validation with a massive reduction in computational latency.

Method	Computational Speed	Boundary Accuracy
Analytical Physics Model	Slow (Sequential trigonometric loops)	Exact Ground Truth
Neural Network Surrogate	Fast (Microsecond batch inference)	$\approx 99.5\%$

Table 5.1: Comparison of real-time feasibility evaluation methods.

Chapter 6

Parametric Design Optimization

6.1 The Generalized Surrogate Model

The kinematic equations demonstrate that L^2 is the leading positive term in the reachability constraints. Thus, increasing the link length L directly delays the onset of internal singularities. To utilize the AI for parametric design, a Generalized Neural Network was trained with three inputs: $(\theta, \phi, L) \rightarrow \text{Feasible}$.

It is critical to note that the neural network does not directly perform the optimization mathematically. Instead, it acts as a high-speed surrogate model for feasibility evaluation. This significantly accelerates the massive parametric sweep required to find the optimal geometry.

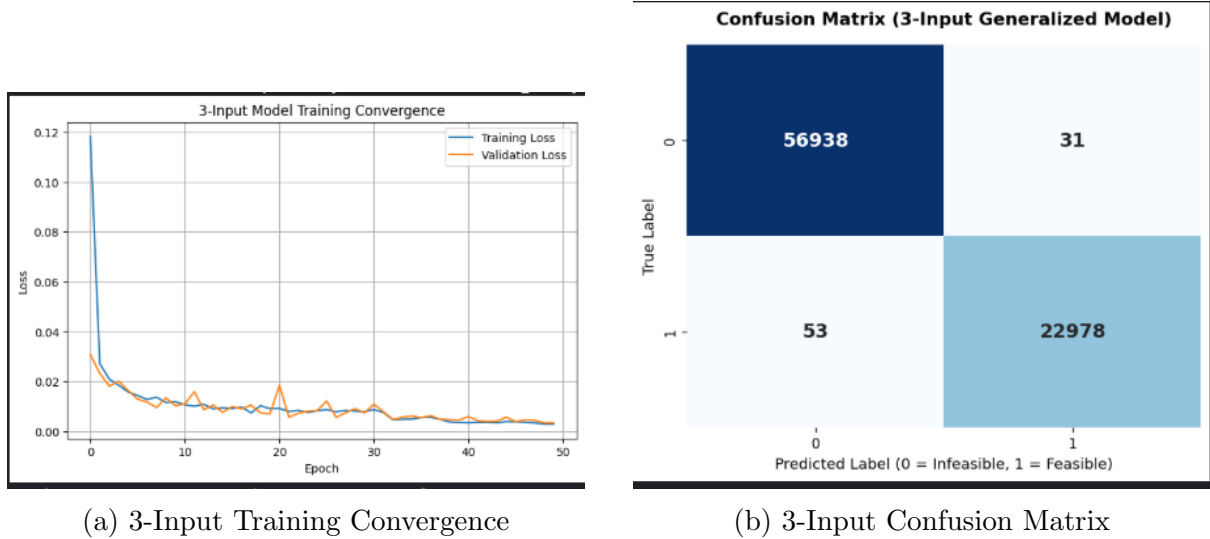


Figure 6.1: Training metrics for the 3-input generalized Neural Network, proving robust generalization across varying manipulator geometries.

6.2 Analytical vs. NN Optimization Sweep

An optimization sweep was conducted, varying L from 10 mm to 75 mm. Both the analytical physics engine and the AI surrogate performed the sweep to calculate total workspace area. The Neural Network’s predicted optimization behavior successfully tracks the analytical ground truth.

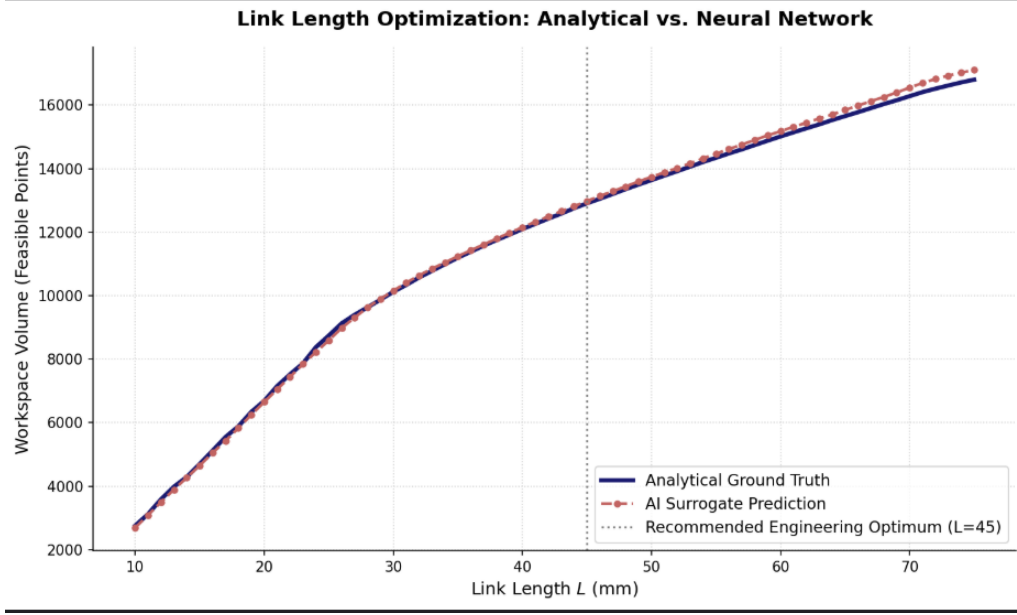


Figure 6.2: Parametric optimization results utilizing the 3-Input Generalized Neural Network. This visualization dictates the physical expansion of the workspace as Link Length L increases.

6.3 Engineering Trade-offs and Final Recommendation

The optimization results reveal a region of extreme diminishing returns:

- **Current Design:** At the original $L = 25$ mm, the workspace is functional but restricted.
- **The Optimum "Knee":** Pushing L to 45 mm results in a massive $\approx 60\%$ expansion of the orientation workspace.
- **The Structural Limit:** While kinematics permit $L > 75$ mm, structural mechanics dictates that once the link length exceeds the base footprint ($S_B = 75$ mm), the legs act as long levers. This results in a cubic loss of bending stiffness ($\delta \propto L^3$) and severe degradation of force transmission angles.

Based on the coupled analytical and AI optimization analysis, the optimal design parameter for this 2PRU-1PUS manipulator is firmly established at $\mathbf{L} = \mathbf{45\ mm}$. This yields maximum orientation flexibility without sacrificing the defining high-stiffness characteristics of parallel manipulators.

Chapter 7

Conclusion

This project successfully achieved its objectives by bridging classical mechanical kinematics with modern machine learning. The complex, highly non-linear singularity limits of the 3-DOF 2PRU-1PUS manipulator were explicitly derived. A Neural Network surrogate model was trained to near-perfect accuracy, completely bypassing the computational burden of real-time trigonometric constraint checking. Finally, the AI model was generalized to act as a high-speed surrogate to assist in the parametric optimization of the manipulator's link length (L). The AI-assisted parametric sweep proved that a minor mechanical adjustment to the link length (from 25 mm to 45 mm) yields a 60% gain in the feasible orientation workspace without sacrificing the defining high-stiffness characteristics of parallel manipulators.